

SASTRA DEEMED TO BE UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

CSE600 / INT600/CSE400 - PROJECT WORK

Project Guidelines

M.Tech/MCA



School of Computing
SASTRA Deemed to be University

TIRUMALAISAMUDRAM
THANJAVUR — 613 401
TAMIL NADU, INDIA

Course Learning Outcome:

On successful completion of the course, students will be able to:

- Define a problem and formulate an engineering problem statement.
- Prepare a methodology for solving the formulated problem, identifying the required parameters and justifying the measures to be used.
- Analyse the data collected and interpret the findings/ solutions.
- Utilize standard tools, algorithms, techniques, software and solve the defined problem.
- Compile a technical report using required representations in the form of graphs, schematic diagrams and charts.
- Work as team player for any engineering activity.

GUIDELINES FOR M.Tech. & M.C.A PROJECT WORK

- The curriculum of M. Tech. degree programme has ‘Project Work and Viva Voce’ (abbreviated as Project) in the 4th Semester for 15 Credits. The learning outcomes for the project are provided in the course directory of the respective programme
- One Credit of project translates to 2 h of work per week
- Project should be individual project, Project can be theoretical, experimental or computational in nature or a combination of these
- Project can be interdisciplinary
- Project shall be one of the following types:

I. In-house: In-house projects are those done at SASTRA (both On–Campus and off-Campus project like design, analysis and simulation type). These students will have a project supervisor from SASTRA. These student(s) need to contact their guides periodically through online/offline mode as applicable.

II. Project Internships other than campus placements: These students typically undertake a project in an industry / institute or research laboratories other than SASTRA with the prior approval of the Dean, SoC. These students will also have a project supervisor from SASTRA for monitoring the progress of the work periodically and ensuring the quality of work. These student(s) need to be in touch with their guides periodically through online.

III. Semester Abroad Program: These are the students who are selected under the Semester Abroad Program of SASTRA. The project supervisor will be from the host institute. These students will not have a project supervisor from SASTRA. Students already selected to carry out project work under SAP may do either online or on-site project work under the supervision of the faculty from the partnering Institution. Such students shall complete the work by first week of June. For them, the CIA will be done by the respective project supervisors abroad.

IV. Project Internships through campus placements: The detailed guidelines shall be communicated soon. As of now, the students under this category are not required to do guide registration on the online project portal.

- All the students of categories I and II will have internal component except SAP students for the project, evaluated at SASTRA by a review panel constituted by the school.
- Thesis submission at the end of the semester and viva voce for 50 marks is common to categories I, II and III.

Evaluation

The assessment of project has the following components:

Continuous Internal Assessment: 50 marks

End Semester Assessment: 50 marks

The total out of 100 will be used for grade as per the Rules & Regulations

- Continuous Internal assessment for projects with supervisor(s) from SASTRA shall comprise of two graded reviews following the scheme as given in the table:

Component		Marks
Review panel	Review I	30
	Review II	30
	Average	30
Project supervisor's component		20
Total		50

Reviews	Evaluation criteria	Marks
Zeroth Review	Not taken for evaluation	NA
	Students will brief on the area or problem identified by them and shall also discuss on the likely objectives for their chosen project to the review panel. The feasibility of the project (considering technical, time and cost factors) will be ascertained. The problem identified / objectives shall be appropriate for project to be executed by a team or as an individual	
Review I	Problem formulation/ Objective & Motivation/ Literature review or survey	10
	Experimental Design / Mathematical Model / Solution Methodology / Methods & Procedures and so on (Methodology / Work plan)	10
	Work carried out	10
	Total	30
Review II	Design/analysis/experiments/fabrication and following of ethical practices & Results	10
	Interpretation/Understanding, Discussion & Conclusion(s) Failed attempts shall also be disclosed and learning from failed attempts or negative results are also acceptable	20
	Total	30

- For student(s) undertaking **SAP** the continuous internal assessment will be carried out by the external guide / on-site mentor. At the end of the project tenure, the project supervisor (external) will fill the feedback form and assess the students based on a 5-point scale for the following six criteria:
 - Fundamental knowledge of the student
 - Independent contribution by the student
 - Willingness to learn new concepts
 - Job knowledge & application skills
 - Personal skills
 - Adherence to organizational rules

Cumulative of ratings for all the 6 criteria (5 points X 6 = 30 points) and will be added to the Project supervisor's component (20 marks), which is explained in Annexure – I. The sum of these two will be taken as internal assessment for the student.

- The scheme of valuation for end semester shall be

Semester Examination	Evaluation criteria	Marks
	Project Report / Thesis	10
	Problem Formulation / Objective & Methods & Processes including experimental design, mathematical model and so on	15
	Results and Discussion	15
	Individual Contribution / Learning and Viva Voce	10
	Total	50

- Faculty panel for Continuous Internal Assessment and for Semester examination shall consist of at least 2 faculty members. Multiple panels shall be formed as per the requirement by the AD/ Dean. One faculty per review panel shall act as coordinator for the panel and shall ensure submission of records to the project coordinator. The AD/ Dean shall constitute the review panel and the project coordinator. The review panel shall be communicated to all students within one week from the date of commencement of the semester by the project coordinator. The students shall be assigned the panel based on the topic or broad area of their work.
- All members of the review panel shall individually assess and award marks to the student(s). The arithmetic mean of the marks awarded by members of the faculty panel shall then be taken as Marks for that review / examination
- During reviews for continuous internal assessment, the faculty members of the panel shall also make specific comments based on the data or material presented for the review. Such comments are also made available to the project supervisor

- ~~The students will have to submit a copy of the project report (spiral bound) adhering to the guidelines to the final exam panel. Any corrections suggested by the panel members should be incorporated and the corrected report (hard bound) should be submitted to the project committee within a week after the viva voce examination~~
- Any student aggrieved with the evaluation of CIA component by the review panel may represent to the respective AD/Dean

Project Diary/Lab note book and Attendance

- Each student shall maintain a project diary/lab note
- The project diary/lab note shall serve as a day-today record of work undertaken by the student towards the project
- The project diary/lab note shall also contain brief minutes of the regular meeting with the project supervisor and the comments given by the project supervisor
- Project diary/lab note should be brought at the time of reviews and also during the semester examination

Schedule for Reviews and Semester Examination

Tentative schedule for review is as follows

- **Zeroth:** Within fortnight from the commencement of the semester
- **First:** Between 5th – 8th week from the commencement of semester
- **Second:** Between 12th - 15th week from the commencement of semester
- **Synopsis and Project Report submission:** 16th week from the commencement of semester
- **Final viva voce:** Dates will be announced by the Controller of Examinations (CoE)

The detailed rubrics for review – I, review – II, project supervisor's component and end-semester examination are provided in Annexure – I.

SYNOPSIS FORMAT

Project Group No:

Register No:

Name:

Write Full Name as per records

1.

2.

3.

Project Title: Project Title to be Typed Using Title Case

Name of the Guide: Name of the guide with affiliation

Abstract

Use this template to prepare the synopsis of the project report. Present here the broad outline of the work that you carried out / focus of the mini project / semester project. These can also include but not restricted to literature survey, objectives identified, experimental and/or modelling methodology adopted, outline of experiments carried out / summary of important results and so on. No need for add any figure or table in this document. No references are required. Use Times New Roman, 12 pt, 1.5" spacing, justify alignment. Maximum no. of words for abstract is 600 and should not exceed ONE A4 sheet. Type the outline as single paragraph. Kindly stick to the format given for submission of synopsis. After the second review meeting, get the signature of the guide and submit the hardcopy to project coordinator. In case of group project each student shall submit an individual synopsis and the synopsis will additionally include the following

Specific Contribution

- *****

Specific Learning

- *****

Technical Limitations & Ethical Challenges faced

- *****

Keywords: Enter 3 – 5 key words that describe your work – Use Times New Roman, 10 pt, Italics

Signature of the Student

Signature of Guide

Date:

GUIDELINES TO PREPARE PROJECT REPORT

PARTS OF PROJECT REPORT

1.	Cover page	Refer Page 18. Format must be adhered to.
2.	Inside cover page	Same as cover Page
3.	Bonafide Certificate	Refer Page 20. Format must be adhered to.
4.	Declaration by Student	Refer Page 21
5.	Certificate (if any, from Industries or other organisations)	
6.	Acknowledgements (Not mandatory)	Not to exceed one page
7.	Abstract	Maximum of 300 words and should contain no more than six key words. Refer Page 26
8.	Table of contents	Refer Page 23
9.	List of Figures	Refer Page 24
10.	List of Tables	Refer Page 25
11.	Abbreviations (if any)	Refer Page 15
12.	Notation (if any)	Refer Page 16
13.	Chapters	Refer Page 27
14.	Appendices	Same format as Chapters
15.	References	Refer Page 29
*	Number of report copies to be submitted (soft copies)	One project report copy to be upload in project portal by any one the team member and synopsis to be uploaded by all the members in a team

STYLE NOTES FOR PROJECT REPORT PREPARATION

Numbering

Pages

Every page in the Project report, except the title page, must be accounted for the page numbering, starting from acknowledgements and till the beginning of the introductory chapter, should be printed in small **Roman numbers, i.e., i, ii, iii, iv, etc.,**

All printed page numbers should be located at the bottom centre of the page, 17 mm (2/3”) from the bottom edge, using normal print.

Chapter

Use only Arabic numerals. Chapter numbering should be centre on the top of the page using large bold print.

Example: **CHAPTER 1**

Sections

Use only Arabic numerals with decimals. Section numbering should be left justified using bold print.

Example:
1.1, 1.2, 1.3, etc.

Equation(s)/ Formula(e)

Use only **Arabic numerals** with single decimal. Equation numbers should be right justified using normal print. Mathematical symbols should be printed in *italics*.

Format: (<Chapter number>.<Equation serial number>)

Example:

$$\mathcal{E} = \left[\frac{2 \cot \left(\phi + \frac{\psi}{2} \right) + \psi \operatorname{cosec} \left(\phi + \frac{\psi}{2} \right)}{\sqrt{3}} \right] \quad (1.1)$$

$$\int_0^{\bar{\varepsilon}_f} \sigma_{\theta} d\bar{\varepsilon} = C_1 \quad (1.2)$$

$$I_1(\bar{\varepsilon}_f) = \frac{2}{3}(2\alpha + 1) \int_0^{\bar{\varepsilon}_f} \bar{\sigma} d\varepsilon_z = \frac{2}{3}(2\alpha + 1) \frac{k(\bar{\varepsilon}_f)^{n+1}}{n+1} \quad (1.3)$$

Please note that the equation numbers are flush right in normal print.

TEXT

Colour: Black print

Font:

Regular text	- Times New Roman 12 pts., and normal print.
CHAPTER HEADINGS	Time New Roman 14 pts. And bold print and all capitals.
SECTION HEADINGS	Time New Roman 12 pts. And bold print and all capitals.
Subsection Headings	Time New Roman 12 pts. And bold print and leading capitals, i.e., only first letter in each word to be the capital.
Special Text	Italics/Superscript/Subscript/Special symbol, etc, as per necessity. Special text may include footnotes, endnotes, physical or chemical symbols, mathematical notations, etc.
References	Same font as regular text. Serial number and all authors names to be in bold print. Journal names and book titles in italics. For format see Annexure 9.

Spacing:

Use **1.15 spacing** between the lines.

Use **a single blank line** between paragraphs.

All paragraphs in the project report should be justified from the first line the last line.

Use **double spacing** between the regular text and quotations.

Provide **one blank line** between:

- (a) Chapter title and first sentence of a Chapter.

Use single line spacing:

- (a) In footnotes and endnotes for text,
- (b) In explanatory notes for tables and figures.
- (c) In text corresponding to bullets, listings, and quotations in the main body of the thesis.

Use **single space** in references and **double space** between references.

Justification: The text should be **fully justified**.

Hyphenation should be avoided as far as possible.

Text corresponding to bullets and listings should be indented.

Quotations from other research work must be indented on the left and the right if they are longer than two lines. Shorter quotations can be included as a part of the regular text.

Windows & Orphans: At the bottom of a page, a paragraph should have at least two lines. Similarly, at the top of a page, a paragraph should end with at least two lines.

MARGINS

The margins for the regular text are as follows:

LEFT	= 1.25"
RIGHT	= 1"
TOP	= 1"
BOTTOM	= 1"

Please note that the bottom of the page numbers should be 17 mm above the bottom edge of the numbered pages.

TABLES

A sample for tables is provided in page 27

All tables should have sharp lines, drawn in black to separate rows/columns as and when necessary.

Tables should follow immediately after they are referred to for the first time in the text. Splitting of paragraphs, to include tables on a page, should be avoided. Provide **three spaces** on the top and the bottom of all tables to separate them from the regular text, wherever applicable.

The last line of the title of any table should be 10 mm to 15 mm **above the top-most horizontal line of the table**, and the title should be centered concerning the table. The titles must be in the same font as the regular text and should be single-spaced. The title format is given below:

Table<blank><chapter number>.<serial number><left indent><table title>.

Example (of a small table which is sought to be placed within the text):

Table 1.1 Results of the simulation experiment.

The contents of the table will be within the surrounding double line (which indicates the top-most, left-most, right-most, and bottom-most boundaries of the table)

Whenever a table exceeds one-page present the full title of the table on the first page and on the following pages, provide the table number and state “(cond.)” after it.

Example: (notice the left justification)

Table 1.2 (contd.)

Wherever explanatory notes are used for clarifying any information presented in the tables, print them after leaving a single space immediately below the tables.

All tables in landscape format must be placed such that their top portions are near the binding of the thesis and their bottom portions near the outer edge.

FIGURES

Sample figure is shown in page 28

All figures, drawings, and graphs should be drawn in black with sharp lines and adequate contrast between different plots if more than one plot is present in the same graph.

Figures should follow immediately after they are referred to for the first time in the text. Splitting of paragraphs, to include figures on a page, should be avoided. Provide three spaces on the top and the bottom of all figures to separate them from the regular text, wherever applicable.

The first line of the title for figures, drawings, graphs and photos should be between 10 mm and 15 mm below the bottom, and they should be centred concerning the figure. The titles must be in the same font as the regular text and should be single-spaced. The title format is given below:

Fig.<blank><chapter number><serial number><left indent><figure title>.

Example:

Fig. 1.1.Stability of the Neuroprosthetics.

Wherever a figure exceeds one page (as in the case of large flow charts for computer programmes) present the full title of the figure on the first page and in the following pages provide the figure number and state “(contd.)” after it.

Example: (notice the left justification)

Fig.1.1 (contd.)

When there are many plots in a single graph or figure, the lettering, labelling or numbering of each plot for its identification should be of a size such that even after size reduction in the report, the identification should be legible.

All Figures in landscape format must be placed such that their top portions are near the binding of the report and their bottom portions near the outer edge.

PHOTOS

Use Colour photos only if they are necessary. Remember that the report may have to be photocopied. In case Colour photos are used, all copies of the report must contain only colour photos.

Photos may be printed on glossy paper, and be mounted with white case in (e.g., Elmer’s glue), glue stick, dry mounting tissue, or any good adhesive. Do not use rubber cement or cello tape.

Each photo should be numbered and referred to as a figure. Photo titles should be similar to those provided for figures.

DRAWINGS

Drawings which are larger than A4 size are not encouraged. If larger drawings are absolutely necessary they may be suitably folded to A4 size in the report.

Each drawing should be numbered and referred to as a figure. Drawing titles should be similar to those provided for figures.

FOOTNOTES

In presenting footnotes and references, use a consistent form acceptable in your discipline.

PUNCTUATION

Please refer to any standard style manual such as the *Chicago Manual of Style*¹, where rules of punctuation are clarified. Student must note that different styles are in practice these days.

¹ *Chicago Manual of Style*, Prentice Hall of India, New Delhi, 1989

For example, some people insert a full stop before ending a sentence in double quotes, whereas other inserts the full stop after the double quotes. Both styles are in practice.

STATISTICAL ANALYSIS

Statistical methods used to analyse the data should be clearly mentioned in the methodology section. Meaning of symbols used in the graphs should be clearly explained in the legends for figures and graphs.

APPENDICES

If the project requires clearances from competent authorities such as Institutional ethics committee (IEC), Institutional Biosafety Committee (IBC), etc., a copy of the approval letter/s should be appended in the appendix.

REFERENCES

References should be arranged in alphabetical order of surname or last name of the first author.

Two or more references by same author(s) in the same year should be indicated by small case alphabets in italics.

For papers by one author:

Bensamaia SJ (2015) Biological and bionic hand: natural neural coding and artificial perception. *Philosophical Transactions of the Royal Society London B Biological Sciences*, **19**, 370-375.

For papers by two authors:

Ghafoor U, Kim S and Hong KS (2017) Selectivity and longevity of peripheral nerve and machine interfaces: A review. *Frontiers in Neurorobotics*, **11**, 59-63.

For papers by more than two authors:

Knutson JS, Fu MJ, Sheffler LR and Chae J (2015) Neuromuscular electrical stimulation for motor restoration in hemiplegia. *Physical Medicine and Rehabilitation Clinics of North America*, **26 (4)**, 729-745.

Adewole DO, Serruya MD, Harris TP, Burrell JC, Petrov D, Chen HI, Wolf JA and Cullen DK (2016) The evolution of neuroprosthetic interfaces. *Critical Reviews in Biomedical Engineering*, **44**, 123-152.

Note: Indicate the names of all authors, if feasible, and avoid the use of *et al* (to indicate the presence of two or more co-authors) after the first author's name.

For books:

Sanchez JC(2016) Neuroprosthetics: Principles and Applications. *CRC Press (USA)*.

For papers presented at Conferences:

Young, E.F. and H.L. Miller (2009) “Old biotechnology to new bio-technology: Continuum or disjunction?” Proceedings of *International Biosymposium for Advancement of Biotechnology*, Tokyo, October, 6-85.

SIMILARITY CHECK REPORT (optional)

- A report on similarity check obtained from Turnitin® shall be submitted along with the project report/ thesis.
- No more than three words in a sentence should be taken from a previous report or publication. Use quotation mark and cite the reference where words and sentences are taken from another work.
- If a figure, table or any data is reproduced from a previous study, prior permission from competent authority should be obtained. Make a mention of the authority from which such permission was obtained.
- References should be cited appropriately when information from previous studies are mentioned in the project report.

ABBREVIATIONS

The student must take utmost care in the use of technical abbreviations. For example, “KM” stands for “Kelvin Mega” and not kilometre (which should be abbreviated as km) and “gms” stands for “gram meter second” and “grams” (which should be abbreviated as g). In addition, abbreviations pertinent to any specific discipline should be listed in alphabetical order as shown below.

AI	Artificial Intelligence
BMI	Brain Machine Interface
CED	Cardiovascular Electronic Device
CNT	Carbon Nano Tube
EPS	Extracellular Polymeric Substances
ICMS	Intracortical Microstimulation
NMES	Neuromuscular Electrical Stimulation
NMJ	Neuromuscular Junction
PNI	Peripheral Nerve Interfaces
QS	Quorum Sensing

NOTATIONS

The student must explain the meaning of special symbols and notations used in the thesis. Define English symbols, Greek symbols, and Miscellaneous symbols separately. Some examples are presented below.

English Symbols (in alphabetical order)

k_d	Microbial decay coefficient
K_s	Substrate concentration when growth rate is half of maximum
bp	base pair
ml	Mililiter
cm	Centimeter
K	Kelvin
Hz	Hertz

Greek Symbols (in alphabetical order)

α	Rate
$\sigma(x)$	the standard deviation of x
Σ	Summation
Ω	Ohm
∞	Infinity

Miscellaneous Symbols (in alphabetical order)

$ x $	absolute value of x
-------	---------------------

TITLE OF THE PROJECT
(14 POINTS, BOLD, TIMES NEW ROMAN, 1.5 LINE SPACE)

*Report submitted to the SASTRA Deemed to be University
in partial fulfillment of the requirements
for the award of the degree of*

Master of Technology/Master of Computer Applications

Submitted by

NAME (12 POINTS, BOLD, TIMES NEW ROMAN)
(Reg. No.: XXXXXXXXXX, Programme)

May 2026



SCHOOL OF COMPUTING

THANJAVUR, TAMIL NADU, INDIA – 613 401



SCHOOL OF COMPUTING
THANJAVUR – 613 401

Bonafide Certificate

This is to certify that the project report titled “**Project Title (Title Case – First Letter Caps)**” submitted in partial fulfillment of the requirements for the award of the degree of **M. Tech. Computer Science and Engineering / M.Tech. Embedded Systems / M.Tech. Cyber Security/ Master of Computer Applications** to the SASTRA Deemed to be University, is a bona-fide record of the work done by **Mr./Ms. AB**(Reg. No. 123456789) (all members name with register no in bracket if applicable) during the final semester of the academic year 2025-26, in the **School of Computing**, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

Signature of Project Supervisor :

Name with Affiliation :

Date :

Project Viva voce held on _____

Examiner 1

Examiner 2



**SCHOOL OF COMPUTING
THANJAVUR – 613 401**

Declaration

I declare that the project report titled “**Project Title (Title Case – First Letter Caps)**” submitted by me is an original work done by me under the guidance of **Dr. Xyzzzzzzz, Designation, School of Computing, SASTRA Deemed to be University** during the final semester of the academic year 2025-26, in the **School of Computing**. The work is original and wherever I have used materials from other sources, I have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

Signature of the candidate(s) :

Name of the candidate(s) :

Date :

Acknowledgements (Sample)

I/We would like to thank our Honorable Chancellor **Prof. R. Sethuraman** for providing me/us with an opportunity and the necessary infrastructure for carrying out this project as a part of our curriculum.

I/We would like to thank our Honorable Vice-Chancellor **Dr. S. Vaidhyasubramaniam** and **Dr. S. Swaminathan**, Dean, Planning & Development, for the encouragement and strategic support at every step of our college life.

I/We extend my/our sincere thanks to **Dr. R. Chandramouli**, Registrar, SASTRA Deemed to be University for providing the opportunity to pursue this project.

I/We extend my/our heartfelt thanks to **Dr. V. S. Shankar Sriram**, Dean, School of Computing, **Dr. R. Muthaiah**, Associate Dean, Research, **Dr. K. Ramkumar**, Associate Dean, Academics, **Dr. D. Manivannan**, Associate Dean, Infrastructure, **Dr. R. Alageswaran**, Associate Dean, Students Welfare.

My/Our guide <Guidename>, <Designation>, School of Computing was the driving force behind this whole idea from the start. His/Her deep insight in the field and invaluable suggestions helped me/us in making progress throughout our project work. I/We also thank the project review panel members for their valuable comments and insights which made this project better.

I/We would like to extend our gratitude to all the teaching and non-teaching faculties of the School of Computing who have either directly or indirectly helped us in the completion of the project.

I/We gratefully acknowledge all the contributions and encouragement from my family and friends resulting in the successful completion of this project. I/We thank you all for providing me an opportunity to showcase my skills through project.

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Abstract

The body should be justified and typed in Times New Roman, 12 pt, not bold, with 1.5 line spacing and should be one page. The abstract should contain a brief background including the literature survey that leads to the research question/objective, methodology and the major results with overall conclusions.

For a team project, each member of the team shall write the abstract in his/her own words. Abstract shall also explicitly state in bullet format the specific contribution and specific learning by each individual. Project thesis / report shall have abstract of all the team members.

Specific Contribution

- *****
- *****
- *****

Specific Learning

- *****
- *****
- *****

CHAPTER 1

INTRODUCTION

Neuroprosthetic devices are in great demand worldwide. Quality of life for patients suffering from neurodegenerative diseases, stroke, brain and spinal cord injuries and limb amputees can be dramatically improved by suitable neuroprosthetic devices. However, there are several challenges that need to be overcome before such devices can be a reality. To be able to design neuroprosthetic devices a multidisciplinary approach involving expertise from basic neuroscience, physiology, medicine, engineering and material sciences will be required. Hence it is imperative that more and more laboratories across the world should focus their efforts to resolve these hurdles and develop neuroprosthetics.

Several approaches have been taken towards the development of neuroprosthetics suitable for specific purpose. Neuromuscular electrical stimulation (NMES) has been used for direct stimulation of muscles in patients having upper motor neuron injuries due to stroke or spinal cord damage (Knutson *et al.* 2015). Peripheral nerve interfaces (PNIs) are being developed for upper-limb amputees (Ghafoor *et al.* 2017). Intracortical microstimulation (ICMS) is used to provide artificial stimulation to the primary somatosensory cortex that mimics the sensation of touch and proprioception (Bensmaia 2015, Tabot *et al.* 2015). Cortical Brain Machine Interfaces (BMIs) help paralyzed subjects to control a robotic arm or wheel chair (Andersen *et al.* 2014, Slutzky and Flint 2017). These devices are summarized in table 1.1 below.

Table 1.1 Neuroprosthetic devices and their applications

Neuroprosthetics	NMES	PNIs	ICMS	BMIs
Application	Upper motor neuron injuries	Upper limb amputees	Peripheral nerve injuries	Paralysis

NMES, neuromuscular electrical stimulation; PNIs, peripheral nerve interfaces; ICMS, intracortical microstimulation; BMIs, brain machine interfaces

Neuroprosthetic interfaces have been designed to facilitate communication between the nervous system (central or peripheral) and a prosthetic device such as a robotic arm or leg (Adewole *et al.* 2016). The design of a neuroprosthetic interface depends on its specific application. However, some of the desired characteristics that all neuroprosthetic interfaces must possess are biocompatibility, high resolution/selectivity and long-term reliability/stability (Adewole *et al.* 2016). In addition, these devices must be affordable for people belonging to poor economic background.

Biocompatibility can be tested in preclinical studies using rodents and monkeys. Cost of these devices can be greatly reduced by using material that is locally available rather than importing them from other countries. Also, one needs to test these devices in the local settings to check the effect of different weather conditions and the specific application for which these devices will be used.

In the present study we have designed neuroprosthetic for patients suffering from peripheral nerve injuries using material available locally. We also tested these devices in rats and monkeys for any immunological reactions, tissue injuries and discomfort. Our preliminary results suggest that these devices can be used for manipulating robotic arms or prosthetic limbs without any obvious discomfort to the subject for at least six months. At the end of six months, the efficiency of these devices is as good as at the beginning of the experiments. Results from one of our tests are shown in Fig. 1.1 below. Further studies to test these devices in clinical settings need to be carried out to examine their suitability for human subjects.

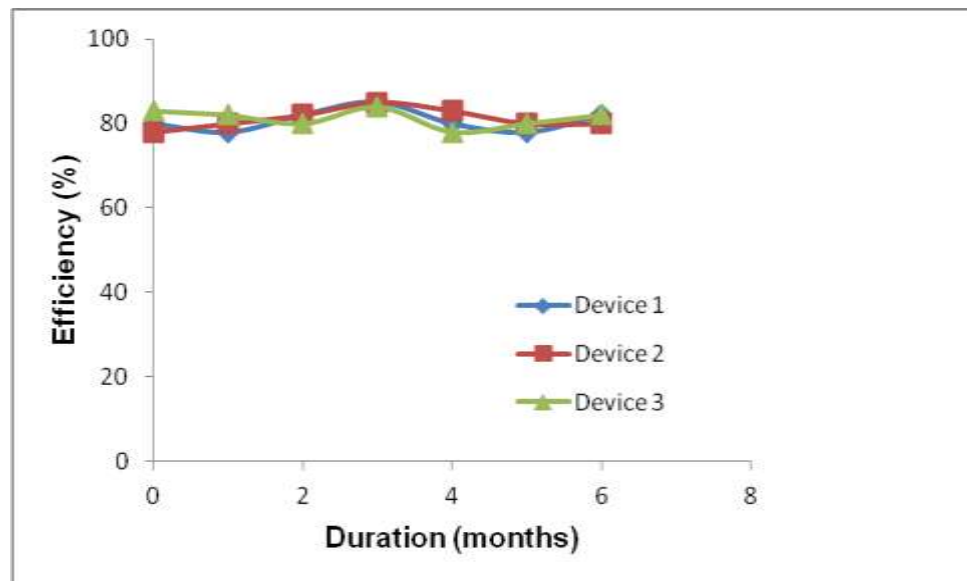


Fig. 1.1 Stability of the neuroprosthetics

REFERENCES

- Adewole DO, Serruya MD, Harris TP, Burrell JC, Petrov D, Chen HI, Wolf JA and Cullen DK (2016) The evolution of neuroprosthetic interfaces. *Critical Reviews in Biomedical Engineering*, 44, 123-152.
- Andersen RA, Kellis S, Klaes C and Aflao T (2014) Toward more versatile and intuitive cortical brain-machine interfaces. *Current Biology*, 24, R885-R897.
- Bensmaia SJ (2015) Biological and bionic hand: natural neural coding and artificial perception. *Philosophical Transactions of the Royal Society London B Biological Sciences*, 19, 370 (1677).
- Ghafoor U, Kim S and Hong KS (2017) Selectivity and longevity of peripheral nerve and machine interfaces: A review. *Frontiers in Neurobotics*, 11, 59.
- Knutson JS, Fu MJ, Sheffler LR and Chae J (2015) Neuromuscular electrical stimulation for motor restoration in hemiplegia. *Physical Medicine and Rehabilitation Clinics of North America*, 26 (4), 729-745.
- Sanchez JC (2016) Neuroprosthetics: Principles and Applications. *CRC Press (USA)*.
- Slutzky MW and Flint RD (2017) Physiological properties of brain-machine interface input signals. *Journal of Neurophysiology*, 118, 1329-1343.
- Tabot GA, Kim SS, Winberry JE and Bensmaia SJ (2015) Restoring tactile and proprioceptive sensation through a brain interface. *Neurobiology of Disease*, 83, 191-198.

Annexure – I

Assessment scheme and the rubrics for award of marks

Review - 1: 30 Marks

I. Problem formulation/ Objective & Motivation/ Literature review or survey (Maximum marks: 10)	
	Marks
(a) Problem statement (5 marks)	
Clear definition and formulation of a problem statement	4-5
Fair definition and formulation of a problem statement	2-3
Poor/ill-defined problem statement	0-1
(b) Literature survey (5 marks)	
Exhaustive literature survey relevant to the problem	4-5
Satisfactory literature survey relevant to the problem	2-3
Inadequate literature survey	0-1
II. Methodology/Work plan (Maximum marks: 10)	
Detailed, well-defined, methodology/ work plan	8-10
Partial/incomplete methodology/ work plan	4-7
Ill-defined or unrealistic work plan	0-3
III. Work carried out (Maximum marks: 10)	
Clear evidence of work done in terms of carrying out analysis/fabrication/case studies/any other appropriate activity	8-10
Some work has been carried out	4-7
Very limited/no work has been carried out	0-3

Review - 2: 30 Marks

IV. Problem formulation/ Objective & Motivation/ Literature review or survey (Maximum marks: 10)	
	Marks
(a) Implementation of methodology (4 marks)	
Systematic implementation	4
Satisfactory implementation	2-3
Poor implementation	0-1
(b) Results (4 marks)	
Excellent presentation of data in terms of graphs, figures, tables & strict adherence to norms	4
Satisfactory presentation of data in terms of graphs, figures, tables & adherence to norms	2-3
Poor presentation of data & adherence to norms	0-1
(c) Adherence to codes, standards & ethical practices (2 marks)	
Full conformity	2
Partial conformity	1
Non-conformity	0
V. Interpretation/Understanding, Discussion & Conclusion(s) (Maximum marks: 20)	

(a) Interpretation (10 marks)	
In-depth analysis of data and interpretation of results	8-10
Satisfactory analysis of data	4-7
Insufficient/poor analysis	0-3
(b) Discussion (5 marks)	
Elaborate discussion with professional etiquette	4-5
Satisfactory discussion	2-3
Poor/insufficient discussion	0-1
(c) Attainment of Objectives/ Inferences/ Conclusions (5 marks)	
Successful attainment of objectives / meaningful inferences & conclusion(s)	4-5
Satisfactory attainment of objectives / reasonably meaningful inferences /conclusion	2-3
Non-attainment of objectives/no meaningful inference/conclusion	0-1

Guide's Evaluation: 20 Marks

VI. Project Report (Maximum marks: 10)	
	Marks
(a) Formatting (2 marks)	
Project report is prepared as per the format; references and citations are adequate	2
Major part of the report is as per the format with acceptable citations	1
Project report is not prepared according to format	0
(b) Technical content (5 marks)	
Report is well-written technically with excellent presentation of data in terms of graphs, figures, tables, discussion, inference/conclusion	4-5
Report is satisfactory with data, discussion and inference/conclusion	2-3
Report is unsatisfactory/poor	0-1
(c) Originality (3 marks)	
Similarity index is less than 10 %	3
Similarity index is between 10 % and 25 %	2 (Revision, if required)
Similarity index is between 25 % and 40 %	1 (Thesis requires re-submission)
Similarity index is greater than 40 %	0 (Thesis requires re-submission)

* While assessing the Similarity index, common phrases, known equations, nomenclature, generic terms and references are to be excluded.

VII(A). Assessment of Individual's Contribution to the Team (Maximum marks: 10)	
	Marks
(a) Synopsis (5 marks)	
Reflective report of the Synopsis correlates with the individual's contribution and learnings	4-5
Reflective report of the Synopsis partially correlates with the individual's contribution and learnings	2-3
Synopsis partially correlates /does not correlate with the individual's contribution and learnings	0-1
(b) Individual's contributions (5 marks)	
Individual's contribution in the execution of the project is evident through project diary/lab note, self and peer evaluation forms	4-5
Individual's contribution in the execution of the project is partly evident through project diary/ lab note, self and peer evaluation forms	2-3
Individual's contribution is not evident	0-1

*VII(B). Assessment of Individual's Learnings from the project (Maximum marks: 10)	
	Marks
(a) Synopsis (5 marks)	
Reflective report of the Synopsis clearly describes the learnings	4-5
Reflective report of the Synopsis partly describes the learnings	2-3
Synopsis poorly describes/does not describe the learnings	0-1
(b) Individual's contributions (5 marks)	
Execution of the project is evident through project diary/ lab note	4-5
Execution of the project is partly evident through project diary/ lab note	2-3
Execution of the project is poor/unsatisfactory	0-1

* - for individual projects only

End Semester Evaluation: 50 Marks

I. Problem Formulation / Objective, Methods & Processes including experimental design, mathematical model and so on (Maximum Marks: 15)	
	Marks
(a) Problem statement/objective (3 marks)	
Clear definition and formulation of a problem statement/objective	4-5
Fair definition and formulation of a problem statement/objective	2-3
Poor/ill-defined problem statement/objective	0-1
(b) Literature survey (4 marks)	
Exhaustive literature survey relevant to the problem	4-5
Satisfactory literature survey relevant to the problem	2-3
Inadequate literature survey	0-1
(c) Methodology/Work plan (8 marks)	
Detailed, well-defined, methodology/ work plan	7-8
Partial/incomplete methodology/ work plan	3-6
Ill-defined or unrealistic work plan	0-2
II. Results and Discussion (Maximum marks: 15)	
(a) Quantum & Quality of Work (2 marks)	
Clear evidence of work done in terms of carrying out analysis/fabrication/case studies/any other appropriate activity	2
Some work has been carried out	1
Very limited/no work has been carried out	0
(b) Results (3 marks)	
Excellent presentation of data in terms of graphs, figures, tables & strict adherence to norms	3
Satisfactory presentation of data in terms of graphs, figures, tables & adherence to norms	1-2
Poor presentation of data & adherence to norms	0
(c) Adherence to codes, standards & ethical practices (2 marks)	
Full conformity	2
Partial conformity	1
Non-conformity	0
(d) Interpretation (6 marks)	
In-depth analysis of data and interpretation of results	5-6
Satisfactory analysis of data	2-4
Insufficient/poor analysis	0-1
(e) Discussion (2 marks)	
Elaborate discussion with professional etiquette	2
Satisfactory discussion	1
Poor/insufficient discussion	0
III. Project Report / Thesis (Maximum marks: 10)	
(a) Formatting (3 marks)	
Project report is prepared as per the format; references and citations are adequate	3
Major part of the report is as per the format with acceptable citations	1-2
Project report is not prepared according to format	0
(b) Technical content (7 marks)	
Report is well-written technically with excellent presentation of data in	6-7

terms of graphs, figures, tables, discussion, inference/conclusion	
Report is satisfactory with data, discussion and inference/conclusion	3-5
Report is unsatisfactory/poor	0-2
IV. Individual Contribution/Learning and Viva Voce (Maximum marks: 10)	
(a) Application of conceptual knowledge (5 marks)	
Application of knowledge is clearly evident	4-5
Limited application of knowledge	1-3
No evidence	0
(b) Viva Voce (5 marks)	
Answered most of the questions of the panel	4-5
Able to answer some of the key questions	2-3
Most of the questions not answered	0-1

Assessment Scheme for Review - I of Project

S. No	Name	Reg. No.	Problem formulation/ Objective & Motivation/ Literature review or survey (10)		Experimental Design / Mathematical Model / Solution Methodology / Methods & Procedures and so on (Methodology / Work plan) (10)	Work carried out (10)	Total (30)	Remarks
			Problem statement (5)	Literature survey (5)				
1								
2								

Assessment Scheme for Review - II of Project

S. No	Name	Reg. No.	Design/analysis/experiments/fabrication and following of ethical process in the same & Results (10)			Interpretation/Understanding, Discussion & Conclusion(s) (20)			Total (30)	Remarks
			Implementation of methodology (4)	Results (4)	Adherence to codes, standards & ethical practices (2)	Interpretation (10)	Discussion (5)	Attainment of Objectives/ Inferences/ Conclusions (5)		
1										

Project Supervisor's Mark (20)

S. No	Name	Reg. No.	Project Report (10)			Assessment of Individual's Contribution to the Team / Assessment of Individual's Learnings from the project (10)		Total (20)	Remarks
			Formatting (2)	Technical content (5)	Originality (3)	Synopsis (5)	Individual's contributions/ Execution of project (5)		
1									
2									

Assessment Scheme for End Semester Examination of Project

S. No	Name	Reg. No	I. Problem Formulation / Objective & Methods & Processes including experimental design, mathematical model and so on (15)	II. Results and Discussion (15)	III. Project Report / Thesis (10)	IV. Individual Contribution / Learning and Viva Voce (10)	Total (50)	Remarks

			I. (a)	I. (b)	I. (c)	II. (a)	II. (b)	II. (c)	II. (d)	II. (e)	III. (a)	III. (b)	IV. (a)	IV. (b)		
1																
2																

I. (a): Problem statement/objective (3)

I. (b): Literature survey (4)

I. (c): Methodology/Work plan (8)

II. (a): Quantum & Quality of Work (2)

II. (b): Results (3)

II. (c): Adherence to codes, standards & ethical practices (2)

II. (d): Interpretation (6)

II. (e): Discussion (2)

III. (a): Formatting (3)

III. (b): Technical content (7)

IV. (a): Application of conceptual knowledge (5)

IV. (b): Viva Voce (5)